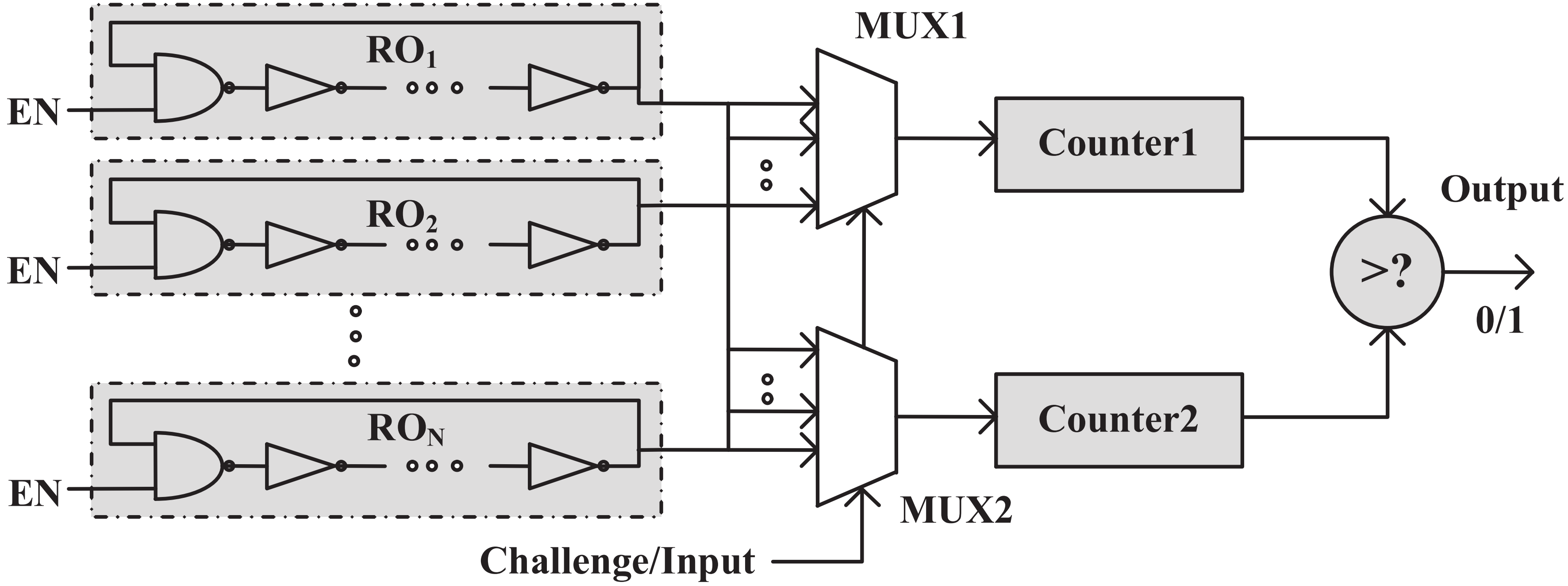
RO PUF



- The structure relies on delay loops and counters instead of MUX and arbiters
- Better results on FPGA – more stable

RO PUF Advantages

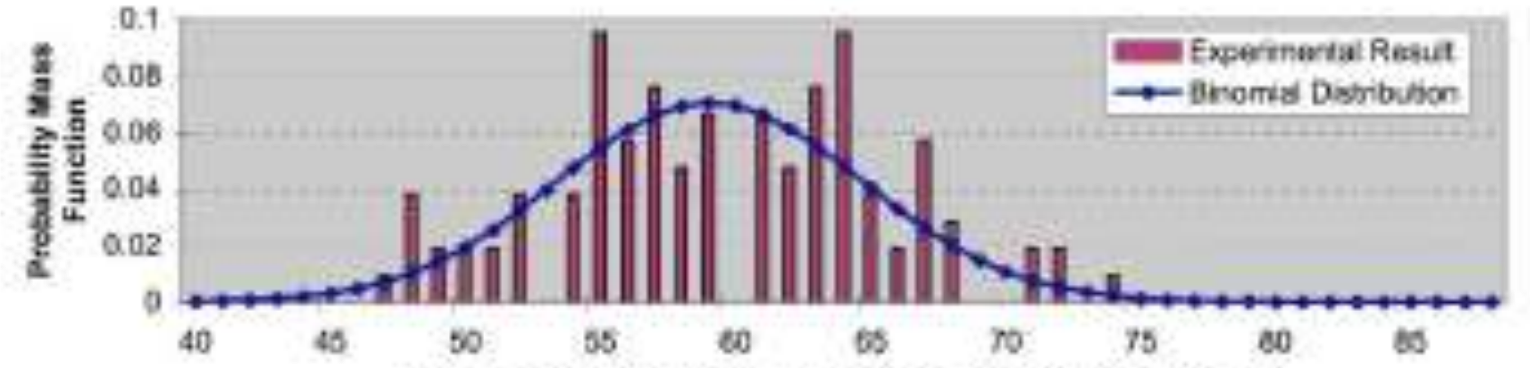
- ROs, whose frequencies are far, are more stable than the ones with closer frequencies
  - Possible advantage: do not use all pairs, but only the stable ones
  - It is easy to watch the distance in the counter and pick the very different ones.
    - Can be done during enrollment
- RO PUF allows an easier implementation for both ASICs and FPGAs
- The Arbiter PUF is appropriate for resource constrained platforms such as RFIDs and the RO PUF is better for use in FPGAs and in secure processor design.

Experiments with RO PUFs

- Experiments done on 15 Xilinx Virtex4 LX25 FPGA (90nm)
- They placed 1024 ROs in each FPGA as a 16-by-64 array
- Each RO consisted of 5 INVs and 1 AND, implemented using lo ok-up tables
- The goal is to know if the PUF outputs are unique (for security) and reproducible (for reliability and security)

Inter-chip Variations

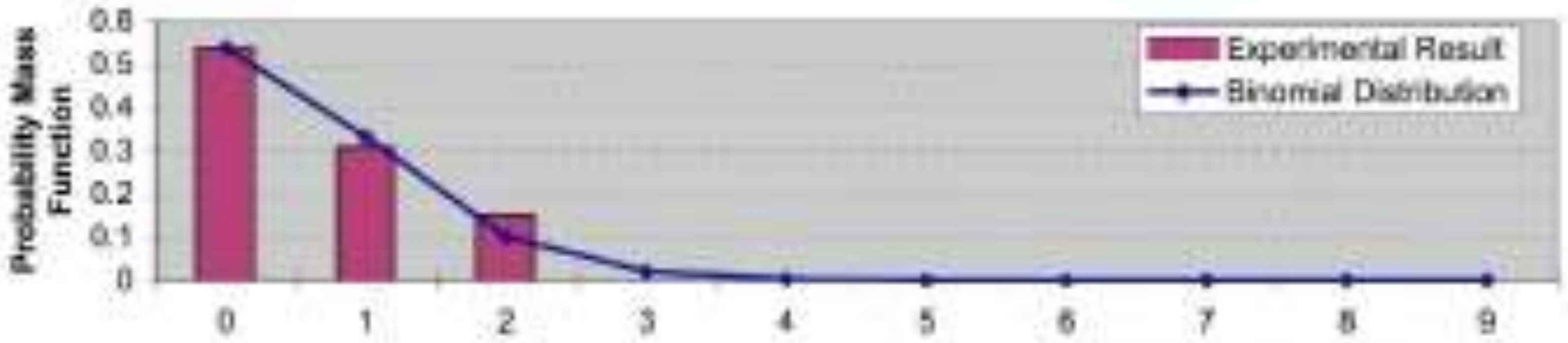
- 128 bits are produced from each PUF
- x-axis: number of PUF o/p bits different b/w two FPGAs; y-axis: probability
- Purple bars show the results from 105 pair-wise comparisons
- Blue lines show a binomial distribution with fitted parameters (n=128, p =0.4615)
- Average inter-chip variations 46.15%



(a) Inter-Chip Variation (Ave = 59.1 bits out of 128 bits, 46.15%)

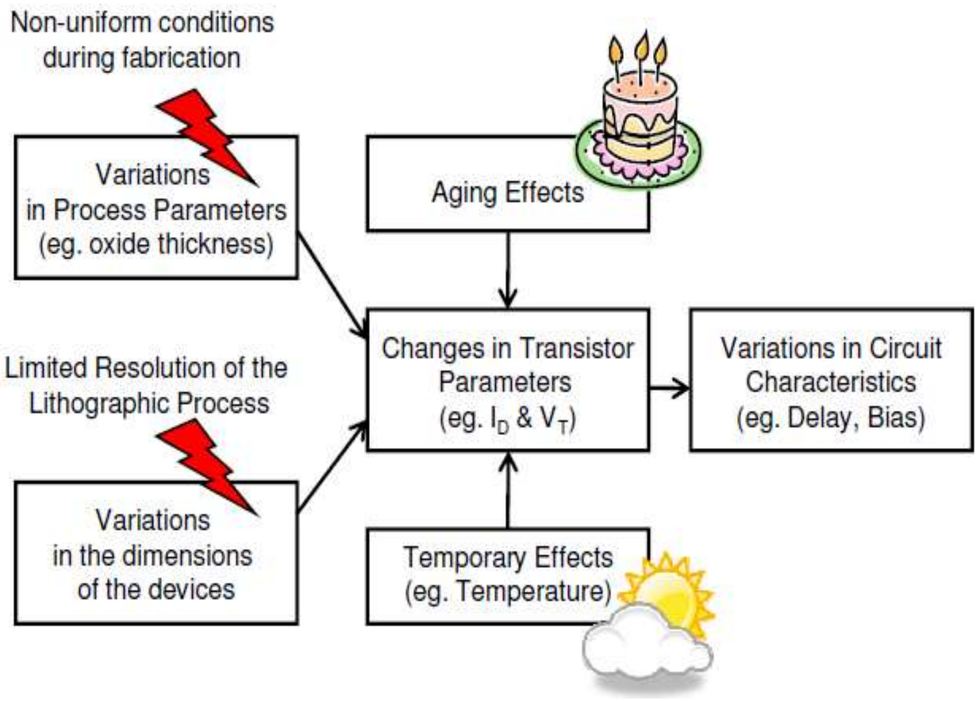
Intra-chip Variations

- PUF responses are generated at two different conditions and compared
- Changing the temperature from 20°C to 120°C and the core voltage from 1.2 to 1.08 altered the PUF o/p by ~0.6 bits (0.47%)
- Intra-chip variations is much lower than inter-chip – the PUF o/p did not change from small to moderate



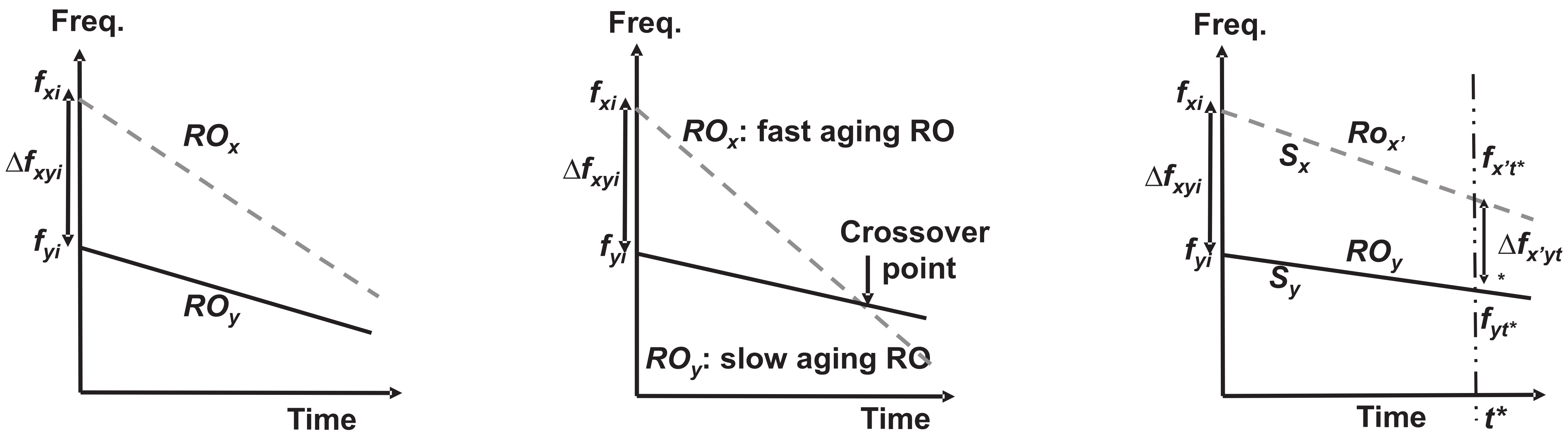
(b) Intra-Chip Var. under the Worst-Case Env. Change (20C 1.2V vs. 120C 1.08V)
(Ave = 0.61 bits out of 128 bits, 0.48%)

Reliability of RO PUFs



- Two types of reliability issues:
- Aging:
  - Negative Bias Temperature Instability
  - Hot Carrier Injection (HCI)
  - Temp Dependent Dielectric Breakdown
  - Interconnect Failure
- Temperature
  - Slows down the device

Bitflip due to Frequency Degradation



(A) Moderate

(B) High degradation

(C) Low degradation

Aging-Resistant RO PUF